

HEAD Completeness for the Weak- L^1 Farkas Recursion

Exact elimination of all higher-dimensional dual freedom in the two-point sector

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Abstract

We isolate and solve exactly the two-point, or *HEAD*, sector of the recursive Farkas separation problem associated with the quartic zero-repair pair

$$P = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = \frac{1}{64}(7, 20, 10, 20, 7).$$

A priori, a new separator may depend on an arbitrary finite collection of lower-horizon separators and on the full coefficient profiles of all of them. We prove that if the resulting separator is supported only at coefficients 0 and 1, then every Farkas certificate collapses to a single scalar mixing parameter θ , one number m recording the smallest lower-horizon right-hand side, and one one-dimensional lower envelope $L(\theta)$ built solely from the projected pairs $(\alpha, d) = (a_1, d)$ of lower-horizon cuts. In particular, every HEAD separator has the exact optimal form

$$\beta(\theta) = \frac{7\theta}{7 + 20\theta}, \quad d_{\text{HEAD}}^{\min}(\theta) = \frac{8(L(\theta) + 20m)}{7 + 20\theta}.$$

For the critical weak- L^1 radius $C = 1/2$, safety is therefore equivalent to the scalar inequality

$$32L(\theta) + 640m \geq 14 + 47\theta.$$

Thus the previously observed FUSION rule is not an ansatz: it is the complete optimal form of every recursively generated HEAD Farkas child. Consequently, every possible HEAD obstruction is reduced to the scalar projected-envelope inequality derived below. Any obstruction not visible to that scalar reduction must come from genuine higher coefficients, i.e. PHASE debt.

1 Exact recursive Farkas setting

Let

$$P = (P_0, \dots, P_4) = \frac{1}{64}(8, 16, 16, 16, 8), \quad Q = (Q_0, \dots, Q_4) = \frac{1}{64}(7, 20, 10, 20, 7).$$

Write

$$F_Q(t) := \sum_{y=0}^t Q_y, \quad F_Q(t) = 0 \quad (t < 0).$$

Thus

$$F_Q(0) = \frac{7}{64}, \quad F_Q(1) = \frac{27}{64}, \quad F_Q(2) = \frac{37}{64}, \quad F_Q(3) = \frac{57}{64}, \quad F_Q(t) = 1 \quad (t \geq 4).$$

At horizon r , a recursive Farkas separator is a monotone finite coefficient vector

$$a = (a_0, a_1, \dots), \quad a_0 \geq a_1 \geq \dots \geq 0,$$

with right-hand side $d \geq 0$, representing

$$\sum_{j \geq 0} a_j b_j \leq d.$$

The exact dual recursion uses multipliers

$$e_x \in \mathbb{Q}, \quad u_k \geq 0 \quad (1 \leq k \leq 4r), \quad v_{xf} \geq 0,$$

where f ranges over lower-horizon separators. The generated coefficients satisfy

$$a_j = \sum_{k=1}^{4r} u_k F_Q(k - j - 1), \quad (1)$$

and the coefficientwise dual-feasibility inequalities are

$$e_x + \sum_{1 \leq k \leq x+j} u_k + \sum_f v_{xf} a_j^{(f)} \geq 0 \quad (x = 0, \dots, 4; j \geq 0). \quad (2)$$

The corresponding raw right-hand side is

$$d_{\text{raw}} = \sum_{x=0}^4 P_x e_x + \sum_{k=1}^{4r} u_k + \sum_{x=0}^4 P_x \sum_f v_{xf} d_f. \quad (3)$$

Only lower-horizon cuts are admitted, so the recursion is acyclic.

A cut with $d \geq a_0$ is trivial on probability laws, because

$$\sum_j a_j b_j \leq a_0 \sum_j b_j = a_0 \leq d.$$

Hence every nontrivial cut may be normalized by $a_0 = 1$. We shall always use this normalization for lower-horizon cuts.

Definition 1 (HEAD separator). *A nontrivial normalized separator is a HEAD separator if*

$$a = (1, \beta, 0, 0, \dots), \quad 0 \leq \beta \leq 1.$$

2 Only two source multipliers can survive

Lemma 1 (Support collapse). *Suppose a recursively generated nonzero separator satisfies*

$$a_j = 0 \quad (j \geq 2).$$

Then

$$\boxed{u_k = 0 \quad (k \geq 3).}$$

Consequently, writing

$$T := u_1 + u_2 > 0, \quad \theta := \frac{u_2}{T} \in [0, 1],$$

one has

$$a_0 = \frac{T(7 + 20\theta)}{64}, \quad a_1 = \frac{7T\theta}{64}. \quad (4)$$

After normalization by a_0 , the HEAD parameter is therefore

$$\boxed{\beta(\theta) = \frac{7\theta}{7 + 20\theta}.} \quad (5)$$

Proof. By (1) with $j = 2$,

$$0 = a_2 = \sum_{k=1}^{4r} u_k F_Q(k-3).$$

Every u_k is nonnegative, and $F_Q(k-3) > 0$ exactly when $k \geq 3$. Hence $u_k = 0$ for all $k \geq 3$.

With only u_1, u_2 left,

$$a_0 = u_1 F_Q(0) + u_2 F_Q(1) = \frac{7u_1 + 27u_2}{64} = \frac{T(7 + 20\theta)}{64},$$

while

$$a_1 = u_2 F_Q(0) = \frac{7T\theta}{64}.$$

Dividing gives (5). □

Remark 1. *This is the first completeness statement: no hidden higher- k source multiplier can contribute to a HEAD child. The entire source side has collapsed to one scalar θ .*

3 Projection of all lower-horizon geometry

For every nontrivial lower-horizon separator f , write

$$\alpha_f := a_1^{(f)}, \quad d_f := \text{its normalized right-hand side.}$$

Define

$$m := \inf_f d_f, \tag{6}$$

and the closed convex projection set

$$\Pi := \text{cl conv}\{(\alpha_f, d_f) : f \text{ is an admissible lower-horizon cut}\}. \tag{7}$$

The closure is harmless and allows us to state exact infima without an attainment assumption.

For $\theta \in [0, 1]$ and $(\alpha, d) \in \Pi$, define

$$\ell_\theta(\alpha, d) := 1 + \inf_{y \geq 0} [yd - \min\{y, 1 - \theta + \alpha y, 1\}], \tag{8}$$

and

$$L(\theta) := \inf_{(\alpha, d) \in \Pi} \ell_\theta(\alpha, d). \tag{9}$$

The meaning of L is simple: it is the exact cheapest normalized lower-horizon extension cost seen by child $x = 0$ once the output is constrained to be HEAD.

Lemma 2 (Closed form of the one-cut extension cost). *Assume $0 \leq \theta \leq 1$, $0 \leq \alpha \leq 1$, and $0 \leq d \leq 1$. Then*

$$\ell_\theta(\alpha, d) = \begin{cases} d, & \alpha \geq \theta, \\ 1 - \frac{1-\theta}{1-\alpha}(1-d), & \alpha < \theta \text{ and } d \geq \alpha, \\ \frac{\theta d}{\alpha}, & 0 < d \leq \alpha < \theta. \end{cases} \tag{10}$$

For $\alpha = 0 < \theta$, the middle formula gives

$$\ell_\theta(0, d) = \theta + (1 - \theta)d.$$

Proof. Let

$$g(y) = yd - \min\{y, 1 - \theta + \alpha y, 1\}.$$

If $\alpha \geq \theta$, then $1 - \theta + \alpha y \geq y$ for $0 \leq y \leq 1$, and the minimum is y on $[0, 1]$ and 1 on $[1, \infty)$. Since $0 \leq d \leq 1$, g is minimized at $y = 1$, giving $\ell_\theta = d$.

Now assume $\alpha < \theta$. For $\alpha > 0$, set

$$y_- := \frac{1 - \theta}{1 - \alpha} < 1, \quad y_+ := \frac{\theta}{\alpha} > 1.$$

Then the minimum in (8) equals respectively

$$y, \quad 1 - \theta + \alpha y, \quad 1$$

on the intervals $[0, y_-]$, $[y_-, y_+]$, and $[y_+, \infty)$. If $d \geq \alpha$, the middle affine piece is nondecreasing, so the optimum occurs at y_- ; if $d \leq \alpha$, it is nonincreasing, so the optimum occurs at y_+ . Substitution yields (10). The case $\alpha = 0$ follows by direct substitution or continuity. $\square$

4 Exact optimization over all Farkas multipliers

Theorem 1 (HEAD completeness). *Fix a lower-horizon cut family and let m , Π , and L be defined by (6)–(9). Among all recursively admissible Farkas certificates whose output is a HEAD separator, the exact optimal normalized right-hand side at source parameter $\theta \in [0, 1]$ is*

$$\boxed{d_{\text{HEAD}}^{\min}(\theta) = \frac{8(L(\theta) + 2\theta m)}{7 + 20\theta}}. \quad (11)$$

The corresponding normalized HEAD coefficient is necessarily

$$\boxed{\beta(\theta) = \frac{7\theta}{7 + 20\theta}}. \quad (12)$$

In particular, no information from the full lower-horizon coefficient profiles $a_2^{(f)}, a_3^{(f)}, \dots$ can improve a HEAD child beyond the two projected data

$$m, \quad \Pi \subset [0, 1] \times [0, 1].$$

Proof. By lemma 1, only u_1, u_2 survive. Write

$$u_1 = T(1 - \theta), \quad u_2 = T\theta.$$

We minimize the contribution of each child independently in (3).

Children $x = 2, 3, 4$. For every $j \geq 0$ one has $x + j \geq 2$, hence

$$\sum_{k < x+j} u_k = T.$$

Choosing no lower-horizon dependency and $e_x = -T$ satisfies (2) with equality somewhere. No smaller e_x is possible because for sufficiently large j all lower-horizon coefficients vanish. Thus the exact normalized contribution of each of these children is

$$-T. \quad (13)$$

Child $x = 1$. Set

$$y_f := \frac{v_{1f}}{T}, \quad Y := \sum_f y_f, \quad E := \frac{e_1}{T}.$$

The only nonredundant constraints are

$$E + 1 - \theta + Y \geq 0, \quad E + 1 \geq 0.$$

Therefore

$$E \geq -\min\{1 - \theta + Y, 1\}.$$

For a fixed total weight Y , the dependency cost is minimized by cuts whose d_f approach m . Hence the optimal child-1 cost divided by T is

$$\inf_{Y \geq 0} [Ym - \min\{1 - \theta + Y, 1\}].$$

For $0 \leq m \leq 1$, the minimum occurs at $Y = \theta$, and equals

$$\theta m - 1. \tag{14}$$

Child $x = 0$. Put

$$y_f := \frac{v_{0f}}{T}, \quad Y := \sum_f y_f, \quad A := \sum_f y_f \alpha_f, \quad D := \sum_f y_f d_f, \quad E := \frac{e_0}{T}.$$

At $j = 0$ and $j = 1$, the exact constraints are

$$E + Y \geq 0, \quad E + 1 - \theta + A \geq 0.$$

For sufficiently large j , every lower-horizon coefficient vanishes, giving $E + 1 \geq 0$. Since all omitted lower-horizon coefficients are nonnegative, these three inequalities are also sufficient for all intermediate j . Thus

$$E \geq -\min\{Y, 1 - \theta + A, 1\}.$$

If $Y = 0$, the cost is 0. If $Y > 0$, write

$$\alpha = \frac{A}{Y}, \quad d = \frac{D}{Y}.$$

Then $(\alpha, d) \in \Pi$, and the normalized child-0 cost is

$$Yd - \min\{Y, 1 - \theta + \alpha Y, 1\}.$$

Taking the infimum over $Y \geq 0$ and $(\alpha, d) \in \Pi$ gives exactly

$$L(\theta) - 1. \tag{15}$$

Conversely, every point of Π can be approximated by convex combinations of admissible lower-horizon cuts, so the same value is attainable as an infimum. Thus no information has been lost.

Assembly. Insert (13), (14), and (15) into (3). Since

$$(P_0, P_1, P_2, P_3, P_4) = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right),$$

we obtain

$$\begin{aligned} d_{\text{raw}} &= T + \frac{1}{8}T(L(\theta) - 1) + \frac{1}{4}T(\theta m - 1) - \frac{1}{4}T - \frac{1}{4}T - \frac{1}{8}T \\ &= \frac{T}{8}(L(\theta) + 2\theta m). \end{aligned}$$

By (4),

$$a_0 = \frac{T(7 + 20\theta)}{64}.$$

Normalizing by a_0 gives (11). The formula for β was proved in lemma 1. $\square$

Remark 2 (Why this is a completeness theorem). *The theorem does not optimize over a prescribed FUSION ansatz. It begins with the full exact Farkas cone: arbitrary nonnegative lower-horizon multipliers on every child, arbitrary lower-horizon cut profiles, and arbitrary admissible equality multipliers e_x . The HEAD support condition itself forces all of that freedom into θ , m , and the projected convex envelope L . Hence FUSION is the normal form of the whole HEAD sector.*

5 Critical weak- L^1 safety

For $0 < C \leq 1$, let B_C denote the exact Pareto-type root with survival

$$\Pr(B_C \geq j) = \frac{C}{j}, \quad j \geq 1.$$

Then

$$\Pr(B_C = 0) = 1 - C, \quad \Pr(B_C = 1) = \frac{C}{2}.$$

Therefore a normalized HEAD separator $(1, \beta, 0, \dots; d)$ accepts B_C precisely when

$$1 - C + \frac{\beta C}{2} \leq d. \tag{16}$$

Corollary 1 (Scalar safety criterion at $C = 1/2$). *Every recursively generated HEAD separator is safe for the critical root $C = 1/2$ if and only if, for every relevant θ ,*

$$\boxed{32L(\theta) + 64\theta m \geq 14 + 47\theta.} \tag{17}$$

Proof. At $C = 1/2$, (16) becomes

$$d \geq \frac{1}{2} + \frac{\beta}{4}.$$

Substitute (5) and (11). Direct simplification gives

$$d_{\text{HEAD}}^{\min}(\theta) - \left(\frac{1}{2} + \frac{\beta(\theta)}{4} \right) = \frac{32L(\theta) + 64\theta m - 14 - 47\theta}{4(7 + 20\theta)}.$$

The denominator is positive, proving the equivalence. $\square$

6 Interpretation: HEAD geometry is complete; PHASE remains

For a general normalized separator

$$a = (1, \alpha, a_2, a_3, \dots),$$

define

$$h(a) := \sum_{j \geq 0} \frac{a_j - a_{j+1}}{j+1}, \quad a_j \rightarrow 0,$$

so that for the critical root $C = 1/2$ its safety slack may be written

$$s := d - 1 + \frac{1}{2}h(a).$$

Introduce the tail-shape coordinate

$$\eta := 1 - \frac{\alpha}{2} - h(a).$$

For the HEAD truncation

$$a^{\text{HEAD}} = (1, \alpha, 0, 0, \dots)$$

one has

$$h(a^{\text{HEAD}}) = 1 - \frac{\alpha}{2},$$

and therefore

$$\boxed{s_{\text{HEAD}} = s + \frac{\eta}{2}.} \tag{18}$$

Equation (18) explains exactly why theorem 1 is not yet the full logarithmic upper-bound theorem. A separator may have a safe HEAD projection while storing a negative safety defect in its higher coefficients. Such a separator carries genuine *PHASE debt*; it cannot be detected from (α, d) alone.

The reduction leaves two sharply separated tasks. First, the actual recursively generated projection set Π must satisfy the scalar HEAD condition (17); the scaling rule from the companion renormalization analysis supplies a closed positive mechanism for a distinguished family, but a dominance/completeness argument is still required to cover every projected recursive cut. Second, after the HEAD projection is controlled, one must rule out debt hidden entirely in the higher coefficients.

Remaining PHASE lemma. Assuming the projected HEAD inequality (17) holds for the relevant recursive projection set, prove that no recursively generated nontrivial separator can have negative critical slack $s < 0$ while its HEAD projection is safe; equivalently, rule out a pure higher-coefficient debt hidden by the term $\eta/2$ in (18).

Thus the high-dimensional part of the completeness problem has been isolated: HEAD geometry is exactly classified by theorem 1, while genuine PHASE debt is the only information not encoded by (α, d) . A full all-horizon proof still requires both the scalar projected-envelope dominance statement and the PHASE lemma. Once both are established, recursive Farkas induction yields uniform $C = 1/2$ weak- L^1 feasibility; the truncated-reserve argument would then give the logarithmic upper bound.

7 Scope

The result proved here is exact and depth-independent, but deliberately narrower than the final theorem. It establishes:

1. complete elimination of all u_k with $k \geq 3$ in every HEAD child;
2. exact reduction of all child-1 lower-horizon information to the scalar m ;
3. exact reduction of all child-0 lower-horizon information to the two-dimensional convex projection Π and the one-dimensional envelope $L(\theta)$;
4. the exact optimal HEAD formulas (5) and (11);
5. the scalar critical safety criterion (17).

It does *not* assert that every recursive separator is HEAD, nor does it by itself prove the all-horizon $O(\log n)$ upper bound. What it proves is a complete normal form for the HEAD sector. To finish the global argument one must still (i) prove the scalar projected-envelope inequality (17) for the full recursively generated projection set and (ii) exclude genuine PHASE debt in the higher coefficients.